

Kuan-Hung Chen

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Education

University of Warwick

MSc in Statistics (with Distinction)

Focus: *Statistical Inference, Data Analysis, Machine Learning, Neural Networks*

Coventry, UK

Sep 2024 — Sep 2025

National Taiwan University

B.A. in Economics & B.A. in Foreign Languages and Literatures

Double Major, Overall GPA: 3.9 / 4.3

Focus: *Econometrics, Game Theory, Security Market Microstructures*

Taipei, Taiwan

Sep 2019 — Jun 2024

Experience

Market Research Intern

KANTAR Worldpanel (now Worldpanel by Numerator)

Taipei, Taiwan

Jun 2023 — Apr 2024

- Engineered 10+ scalable Python and VBA data pipelines to accelerate data process workflow, making Excel formatting and PowerPoint report production at least 20x faster.
- Analyzed consumer datasets to extract actionable insights for 10+ FMCG clients, delivering reports and presentations that provided market insights and trends.
- Managed the end-to-end business solution lifecycle for a high market-cap, industry-leading client account, delivering data-driven consultancy services.

Research and Projects

MSc Dissertation (University of Warwick)

Probabilistic Weather Forecasting with Recursive Algorithms and Vine Copulas

Sep 2025

- Engineered a vine copula-based neural network for mid-term probabilistic weather forecasting utilizing PyTorch.
- Implemented a Quasi-Bayesian recursive density estimation algorithm to enable data augmentation without re-training, optimizing training efficiency.
- The framework proved stable and high accuracy in chaotic time series datasets as well as real-world WeatherBench datasets, demonstrating potential of practical application.

Open-Source Developer | Intaglio

Obsidian-to-Jekyll Static Content Pipeline

2026

- Engineered a Python pipeline converting Obsidian Markdown into Jekyll-compatible output, implementing incremental builds and asset syncing to ensure posts are up-to-date.
- Hardened the converter against real-world input with a 129-test suite spanning unit, integration, and subprocess-driven CLI test.
- Packaged the tool as a distributable composite GitHub Action exposing 12 inputs and 4 workflow outputs, publishing to a separate site repository through reviewable pull requests.

Machine Learning Project | NLP Product Categorizer

NLP Product Categorizer for zh-TW E-Commerce Context

2026

- Engineered and open-sourced an NLP categorization framework utilizing TF-IDF vectorizer and SVC/MNB classifiers, specifically optimized for the complex naming conventions of the zh-TW e-commerce market (mixing English, numeric, and Chinese characters).
- Developed and containerized a production-ready backend API using FastAPI and Docker, enabling seamless deployment and real-time model inference.
- Achieved 70% accuracy and F1 scores against highly noisy, error-prone real-world datasets, establishing a robust performance baseline.

Machine Learning Project | ALBERT-based Transformer Parser

NLP Poll String Tagger/Parser for zh-TW Natural Phrasing Habits

2026

- Developed an open-source NLP parser and training framework for Traditional Chinese poll text, specifically optimized for the natural phrasing habits of Taiwanese users.
- Engineered a transformer fine-tuning pipeline utilizing a pretrained ALBERT encoder for span-boundary prediction.
- Designed a rule layer utilizing regex-driven grammar and lexicon lookups to convert labelled text spans into structured JSON payloads, which currently powers the backend API of a live polling web app **Sootow**.

Skills

Programming: SQL, Python, C++, R, VBA, JavaScript, Ruby

Machine Learning: Numpy, Pandas, scikit-learn, PyTorch, NLP

Data Analysis: Stata, Tableau, Time Series Forecasting, ETL pipeline

Web Development: Ruby on Rails, React, FastAPI, HTML, CSS

Tools: Bash, Docker, Git, $\text{\LaTeX}$, Excel, PowerPoint

Languages: Mandarin (Native), English (Professional; IELTS Academic 8.0), German (Intermediate)